

Project Development Phase

Date	9 July 2026
Team ID	
Project Name	Food Ordering And Customer Trends
Maximum Marks	4 Marks

Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the Food Ordering Behaviour dataset into Tableau Public.
- Verified that all records and fields (Order ID, Customer Age Group, Cuisine, Meal Type, City, Revenue, Customer Rating, Delivery Time, Restaurant Segment, etc.) were imported successfully.
- Confirmed that the dataset was ready for preprocessing.

Step 2: Check for Missing Values

- Examined the dataset for null or blank values.
- Verified that important fields such as Revenue, Customer Rating, Cuisine, Meal Type, and City contained valid information.
- Missing values were handled appropriately to ensure accurate analysis.

Step 3: Remove Duplicate Records

- Checked the dataset for duplicate order records using Order ID.
- Removed duplicate entries to prevent inaccurate counts and incorrect business insights.
- Ensured that every order was represented only once.

Step 4: Validate Data Types

- Order ID → String
- Revenue → Number
- Customer Rating → Number (Decimal)
- Delivery Time → Number
- Customer Age Group → Categorical
- Cuisine → Categorical
- Meal Type → Categorical
- Restaurant Segment → Categorical
- City → Geographic / String

Step 5: Verify Data Consistency

- Checked for invalid or unrealistic values such as negative revenue, invalid customer ratings, or incorrect delivery times.
- Standardized cuisine names, city names, and restaurant segment values.
- Ensured consistency across all categorical fields before visualization.

Step 6: Prepare Calculated Fields

- Total Orders
- Total Transaction Amount
- Average Customer Rating
- Average Delivery Time
- Transaction Amount by Cuisine / City
- Order Count by Meal Type / Company Type / Age Group
- KPI Measures for Dashboard

Step 7: Final Clean Dataset

- Verified that the cleaned dataset was complete and suitable for analysis.
- Confirmed that all preprocessing activities were completed successfully.
- The dataset was finalized for dashboard creation and business intelligence visualizations in Tableau Public.

Business Questions with Visualization

S.No	Visualization	Professional Business Question
1	KPI Cards	What are the key business performance metrics, including total orders, total transaction amount, average customer rating, and average delivery time?
2	Heat Map (Flavours Across Cities)	How do cuisine preferences vary across different cities based on total transaction amount?
3	Grouped Bar Chart	How does order volume vary across different company types and meal types?
4	Donut Chart	What is the percentage contribution of each meal type to the overall transaction value?
5	Tree Map	Which cities contribute the highest order volume, and how are orders distributed across cities?
6	Histogram	What is the distribution of delivery time across customer orders?
7	Horizontal Bar Chart	How do average customer ratings differ across various meal types?
8	Stacked Bar Chart (Age Group & Restaurant Type)	How are customer orders distributed across different age groups and restaurant segments?
9	Pie Chart (Cuisine Preference)	What is the distribution of customer preferences across different cuisine categories?

Visualizations

Each business question below is paired with the Tableau visualization created to answer it.

1. KPI Cards

Business Question: What are the key business performance metrics, including total orders, total transaction amount, average customer rating, and average delivery time?

Key Performace Matrix			
Total Orders	Total Transaction Amount	Avg. Rating Given	Avg. Time Taken To Order
50,000	30.37M	3	7.50Min.

Key Performance Matrix - Total Orders: 50,000 | Total Transaction Amount: 30.37M | Avg. Rating Given: 3 | Avg. Time Taken To Order: 7.50 Min.

2. Heat Map (Flavours Across Cities)

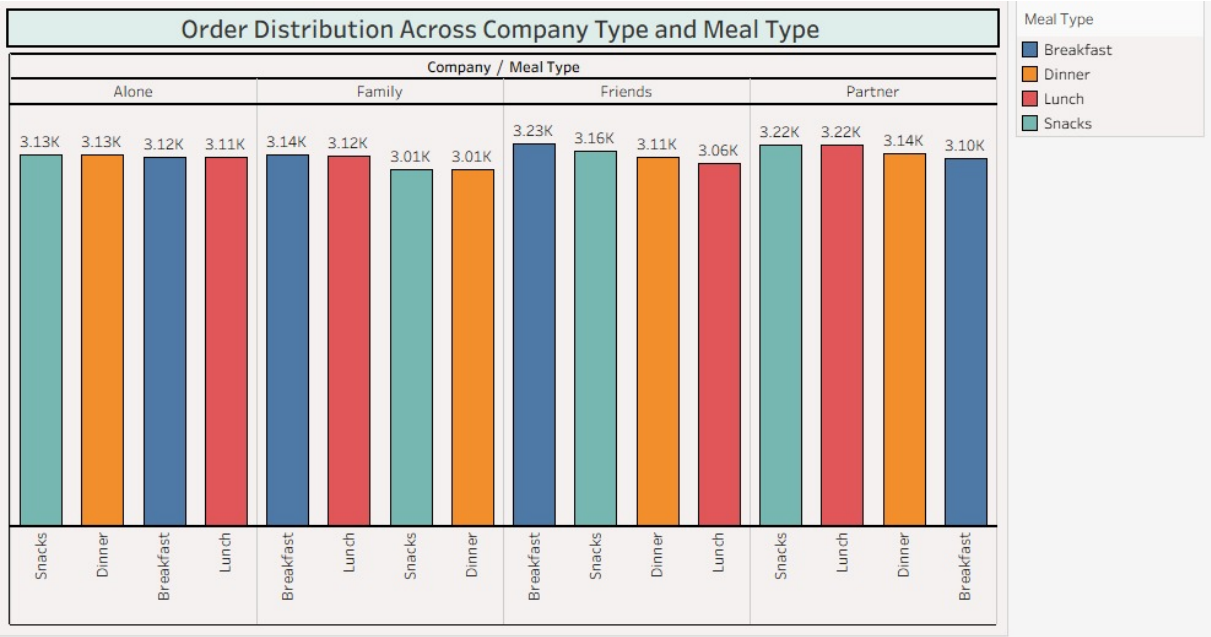
Business Question: How do cuisine preferences vary across different cities based on total transaction amount?

Flavors Across Cities							SUM(Order Value)
City	Cuisine						719.81K 813.99K
	Biryani	Chinese	Desserts	Fast Food	North Indian	South Indian	
Bangalore	740.85K	763.50K	779.99K	793.32K	742.19K	719.81K	
Chandigarh	753.16K	769.98K	750.84K	761.72K	725.86K	764.07K	
Delhi	769.45K	724.41K	778.48K	752.05K	737.23K	775.12K	
Hyderabad	735.47K	767.06K	739.02K	739.60K	765.16K	781.70K	
Mumbai	777.19K	758.85K	813.99K	777.37K	745.65K	767.27K	
Pune	759.66K	797.44K	764.88K	768.71K	750.71K	774.81K	

Flavors Across Cities - transaction value by cuisine and city.

3. Grouped Bar Chart

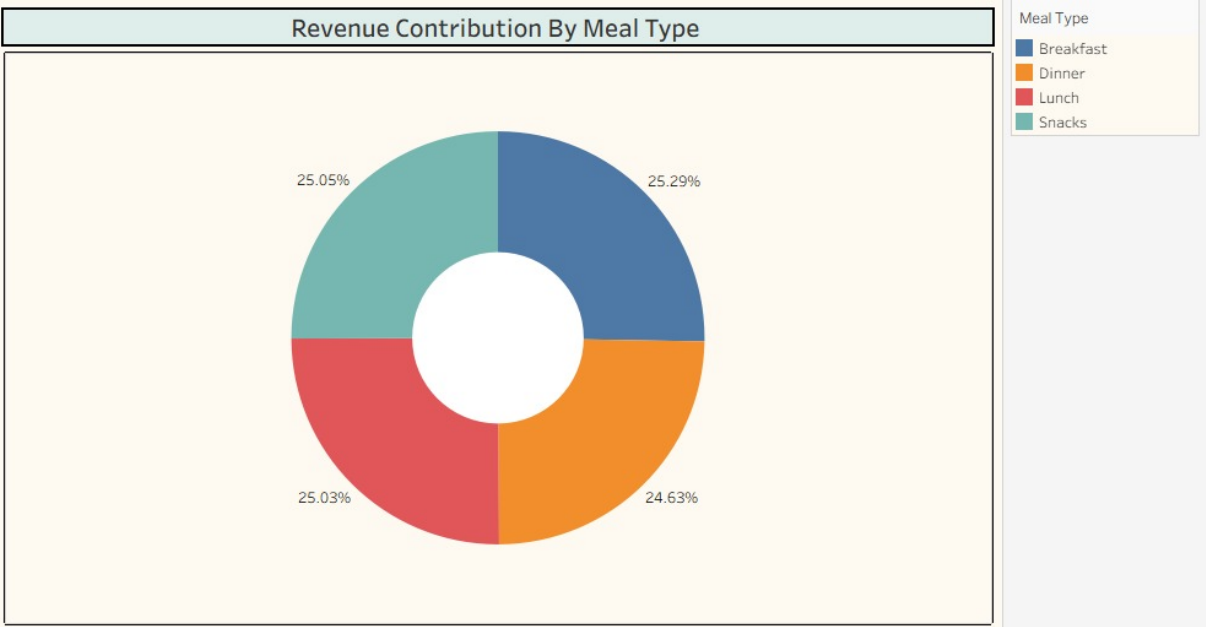
Business Question: How does order volume vary across different company types and meal types?



Order Distribution Across Company Type and Meal Type.

4. Donut Chart

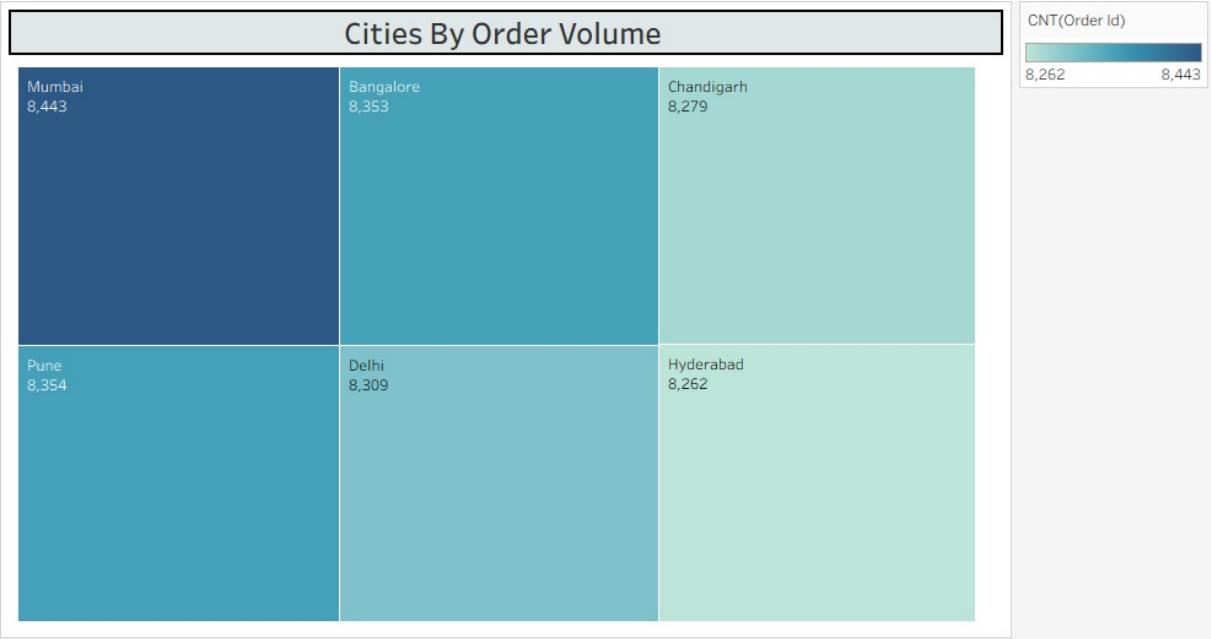
Business Question: What is the percentage contribution of each meal type to the overall transaction value?



Revenue Contribution By Meal Type.

5. Tree Map

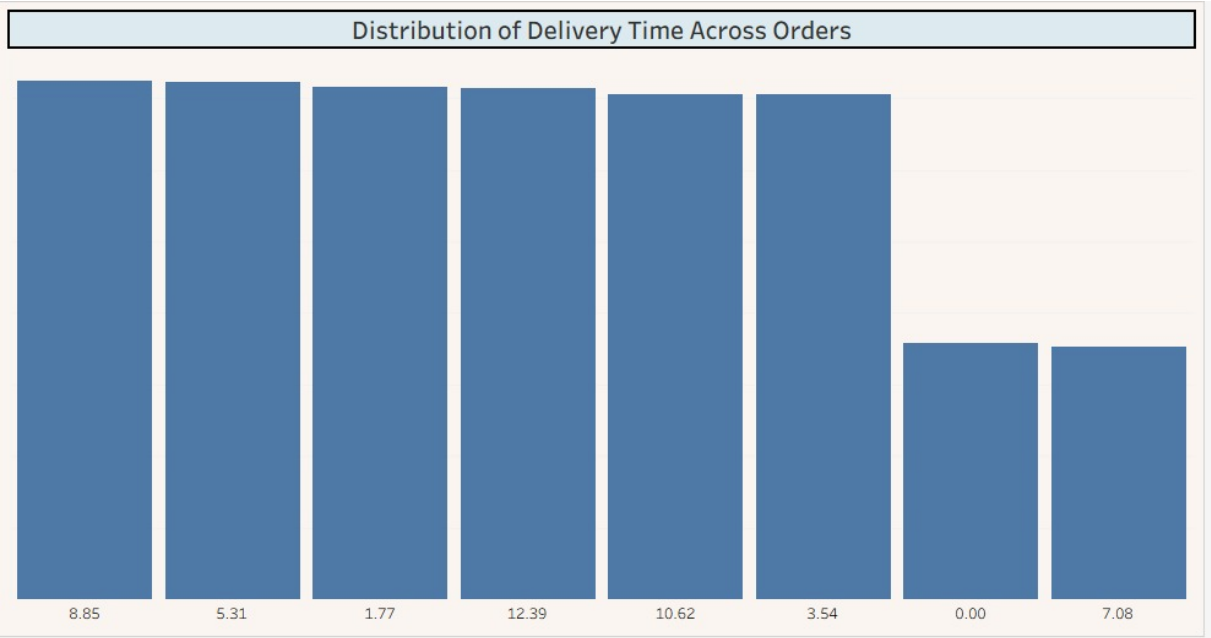
Business Question: Which cities contribute the highest order volume, and how are orders distributed across cities?



Cities By Order Volume.

6. Histogram

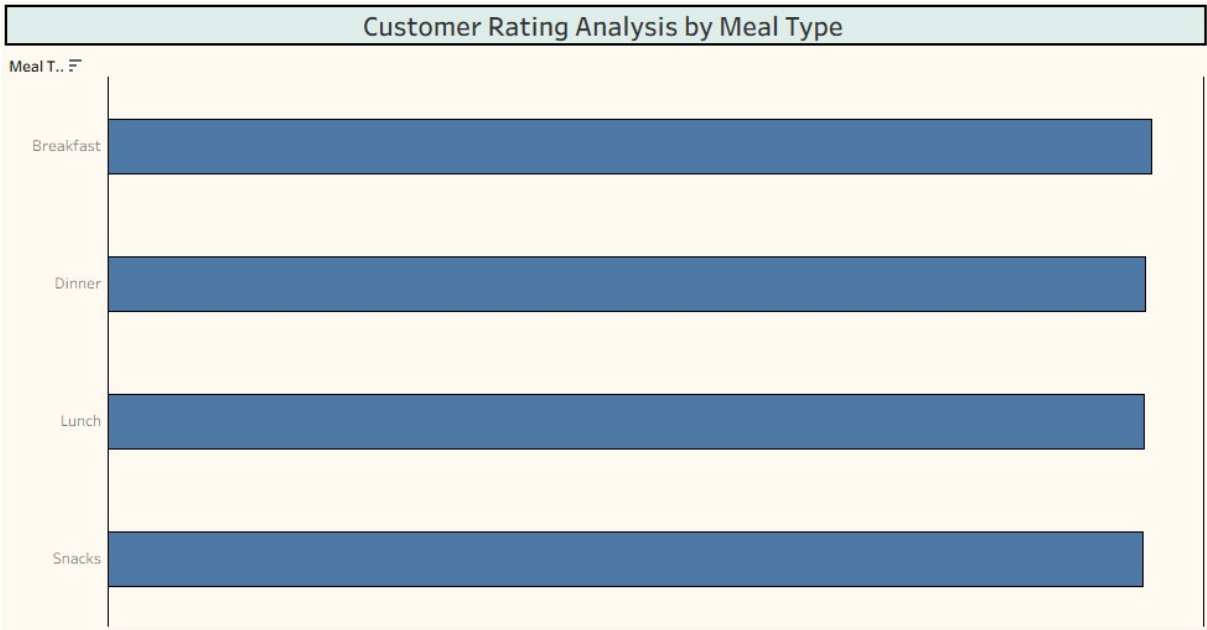
Business Question: What is the distribution of delivery time across customer orders?



Distribution of Delivery Time Across Orders.

7. Horizontal Bar Chart

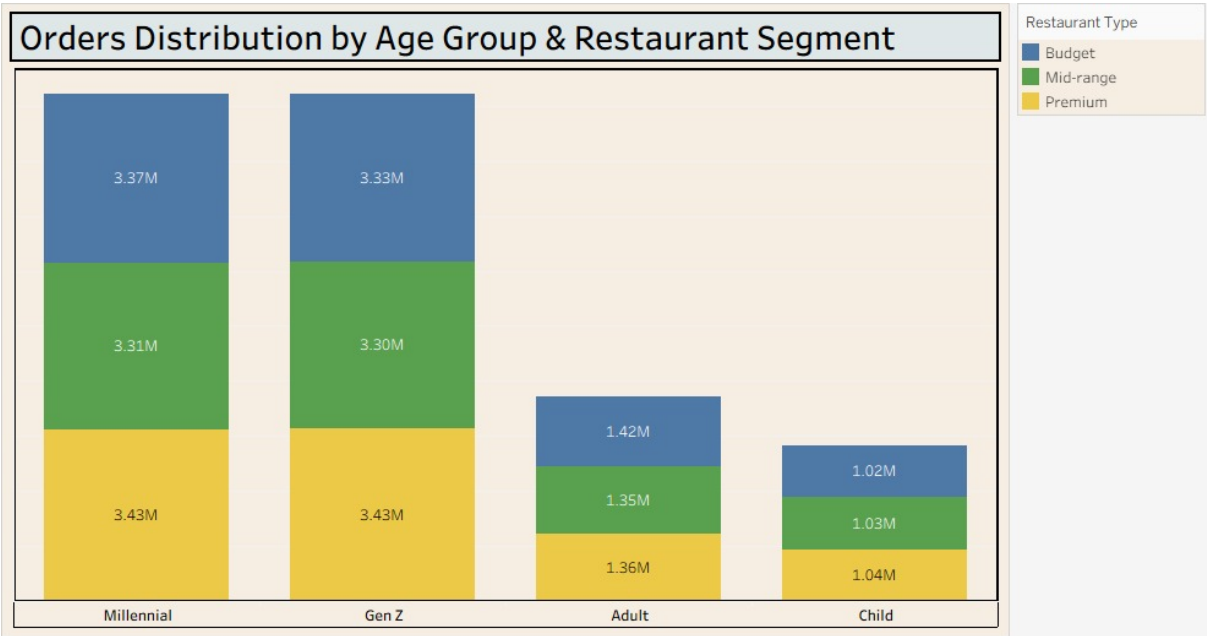
Business Question: How do average customer ratings differ across various meal types?



Customer Rating Analysis by Meal Type.

8. Stacked Bar Chart (Age Group & Restaurant Type)

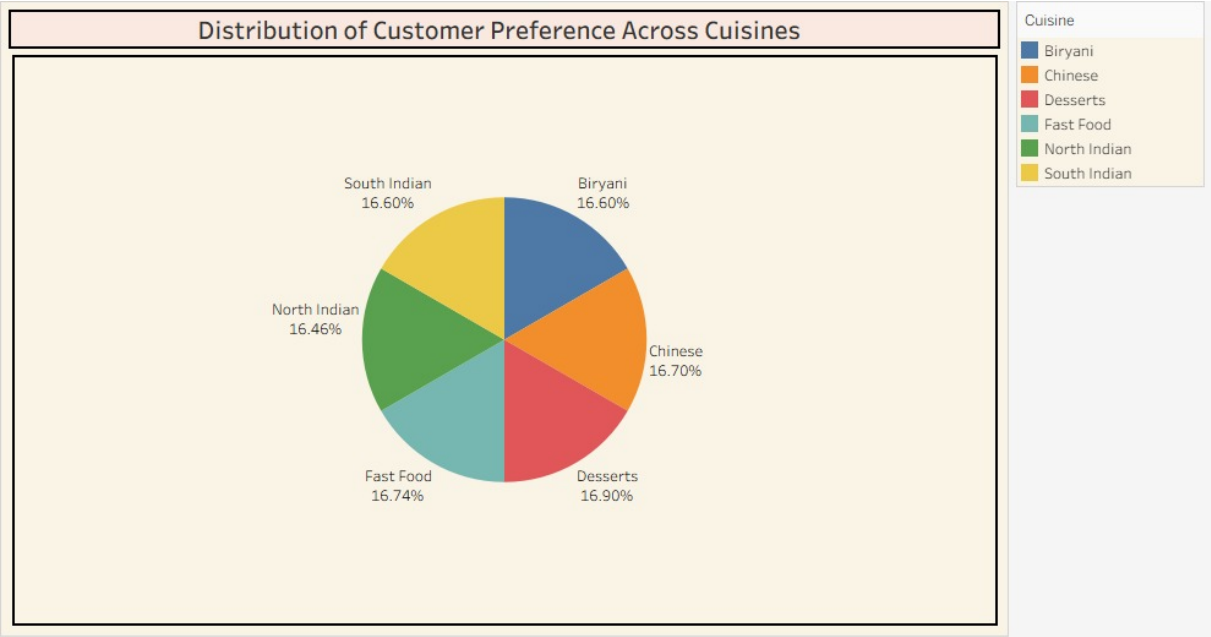
Business Question: How are customer orders distributed across different age groups and restaurant segments?



Orders Distribution by Age Group & Restaurant Segment.

9. Pie Chart (Cuisine Preference)

Business Question: What is the distribution of customer preferences across different cuisine categories?



Distribution of Customer Preference Across Cuisines.